WORKSHEET

Chapter 1: Scalar product

1 In each of the following cases the magnitudes of two vectors and the angle between them is given. Find the scalar product in each case.

a 2, 8, 76°

b 4, 5, 30°

c 14, 3, 90°

d 12, 6, 0°

e 13, 4, 60°

f 15, 10, 240°

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2 Calculate the following scalar products.

a $(4, 110°) \cdot (15, 22°)$

b $(7, 155°) \cdot (12, 210°)$

c $(11, 250°) \cdot (13, 266°)$

d $(5, 130°, 28°) \cdot (7, 187°, 67°)$

e $(12, 217°, 77°) \cdot (16, 110°, -36°)$

f $(16, 35°, -64°) \cdot (5, 23°, 88°)$

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3 Find the following scalar products.

a $(1, 4, 7) \cdot (6, 2, 4)$

b $(5, -3, 1) \cdot (4, -2, -7)$

c $(0, 12, -5) \cdot (3, 0, -2)$

d $\begin{pmatrix} 2 \\ 7 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 4 \\ 8 \end{pmatrix}$

e $\begin{pmatrix} 6 \\ -4 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 5 \\ 7 \end{pmatrix}$

f $\begin{pmatrix} 7 \\ 4 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix}$

4 Given that $\mathbf{a} = -3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = 5\mathbf{i} + -2\mathbf{j} + 7\mathbf{k}$, $\mathbf{c} = \mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$ and $c = 4$, show that:

a $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|$

b $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$

c $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$

d $(c\mathbf{a}) \cdot \mathbf{b} = c(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (c\mathbf{b})$

e $\mathbf{0} \cdot \mathbf{a} = \mathbf{0}$

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5 For $\mathbf{a} = 5\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$, show that $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

6 Find the angle between the vectors.

a $\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$

b $2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ and $3\mathbf{i} + 7\mathbf{j} - 4\mathbf{k}$

c $-3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$ and $9\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$

d $4\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $-5\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}$

e $5\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$ and $3\mathbf{i} + 4\mathbf{j} + \mathbf{k}$

f $-6\mathbf{i} + 10\mathbf{j} + 5\mathbf{k}$ and $7\mathbf{i} - 3\mathbf{j} - 4\mathbf{k}$

7 If $\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + 7\mathbf{j} + 2\mathbf{k}$ and $\mathbf{c} = 4\mathbf{i} - 5\mathbf{j} + 3\mathbf{k}$, find:

a the projection of $\mathbf{a}$ on $\mathbf{b}$

b the projection of $\mathbf{c}$ on $\mathbf{a}$

c the projection of $\mathbf{b}$ on $\mathbf{c}$

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d the projection of $\mathbf{c}$ on $\mathbf{c}$

e the projection of $(\mathbf{a} + \mathbf{c})$ on $\mathbf{b}$

f the projection of $(\mathbf{a} + \mathbf{b})$ on $\mathbf{c}$

8 In each of the following, calculate the work done by the force $\mathbf{u}$ in moving an object from P to Q. The force is measured in newtons and the displacement in metres.

a $\mathbf{u} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$, $P(1, -2, 4)$ and $Q(2, -4, 3)$

b $\mathbf{u} = -\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$, $P(3, -4, 3)$ and $Q(4, -3, 0)$

c $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $P(-4, 6, -3)$ and $Q(7, 5, -2)$

d $\mathbf{u} = -4\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}$, $P(2, -7, 5)$ and $Q(-3, -4, 3)$

e $\mathbf{u} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $P(5, 2, 1)$ and $Q(5, 1, 4)$

f $\mathbf{u} = 7\mathbf{i} + 4\mathbf{j} + 8\mathbf{k}$, $P(-8, 3, 8)$ and $Q(6, 2, 4)$

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Answers

1 **a** 3.87
b 17.32
c 0
d 72
e 26
f −75

2 **a** 2.09
b 48.18
c 137.46
d 60.94
e 67.09
f −69.07

3 **a** 42
b 19
c 10
d 14
e −11
f −4

4–5 Demonstrations

6 **a** 69.1°
b 1.9°
c 154.6°
d 111.2°
e 82.9°
f 147.4°

7 **a** 0.64
b −1.96
c −2.40
d 7.07
e −1.52
f −3.68

8 **a** 4
b 10
c 28
d 31
e 4
f 62

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